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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	Hands Off Our Rights
Project / Solution Name	Future Skills: find the part of IT that suits you, and the job to aim for
Name (Leader)	Hong Zhi Yang
Email & Phone (Leader)	zhiyang19015@gmail.com · +6011-40359109
University Affiliation(s)	Universiti Tunku Abdul Rahman (UTAR)
Live Prototype URL	https://future-skills.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/pekoremon/future-skills

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

The browser front end (plain HTML, CSS and JavaScript, no build step) guides a student through four steps: an optional CV upload or skill ratings, an interest questionnaire, a short skills test on the 3 to 6 IT areas their CV and interests point to, and jobs grouped by interest and skill, with a learning path. Code answers run in the browser with Pyodide; PDF CVs are read with PDF.js.

The back end is one Python 3.12 program using only the standard library. It scores the questionnaire and test, matches the student to 26 IT roles in 6 IT areas and builds learning paths from CSV tables in data/ (O*NET, DOSM, MyCOL, arXiv, course links), where every row carries its source and date. SQL answers run in a read-only, in-memory SQLite database.

Only two features call an AI model: "Explain this for me" and the CV reader (summary and evidence per IT area) send computed facts or CV text to gpt-oss-120b on Fireworks AI; if the call fails, a rule-based text is shown. An "Update data" job refreshes trend inputs from the OpenDOSM and arXiv APIs. The app is hosted on Replit Deployments; the code is on GitHub.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.12 standard library (http.server, sqlite3, csv, json, urllib, unittest)	Back end, data scripts and 140 automated tests; no third-party Python packages	PSF License
JavaScript (ES2020), HTML5, CSS3	Web front end, written by the team without a framework	Open web standards
Pyodide 0.26.4 (loaded from the jsDelivr CDN)	Runs students' Python code answers in the browser, so no code runs on our server	MPL 2.0
PDF.js 3.11.174 (Mozilla, loaded from the cdnjs CDN)	Reads the text of a PDF CV in the browser	Apache 2.0

Name & Version	Purpose in Project	License
SQLite (bundled with Python)	Read-only, in-memory database that runs SQL test answers	Public domain

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Fireworks AI chat completions API (Fireworks AI)	Writes the "Explain this for me" text and the CV summary with gpt-oss-120b	Paid (team's own account)
Data Catalogue API, OpenDOSM / data.gov.my (Department of Statistics Malaysia)	Employment and wage data for the "Update data" trend refresh	Free, no key
arXiv API (arXiv, Cornell University)	Counts AI research papers per IT area for the AI-momentum signal	Free, no key
jsDelivr and cdnjs (public CDNs)	Serve the Pyodide and PDF.js libraries to the browser	Free
Microsoft Learn Catalog API and YouTube oEmbed (development only, not called by the app)	Collecting course titles and lengths, and checking that every learning link exists	Free, no key

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Deployments (Replit)	Hosts the live prototype; the API key is stored in Replit Secrets	Replit Cloud Run deployment target, Replit default region; paid with SDC sponsor credits
Replit Workspace (Replit)	Shared development and test environment	SDC sponsor credits

Service & Provider	Purpose in Project	Plan / Region
GitHub (GitHub, Inc.)	Code repository and commit history	Free plan
Fireworks AI serverless inference (Fireworks AI)	Runs the gpt-oss-120b model the app calls	Pay as you go, team's own account
Replit web analytics (Replit)	Page-view statistics that Replit Deployments adds to the live site	Included with Replit Deployments
Database, file storage, login	None: data ships as CSV files in the repository and nothing is stored per user	Not applicable

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
gpt-oss-120b (OpenAI open-weight model, 120B parameters)	Fireworks AI (accounts/fireworks/models/gpt-oss-120b)	Pre-trained, used as is (no fine-tuning)	Apache 2.0 with the gpt-oss usage policy; Fireworks AI terms of service
No other model	Matching, test grading, trend and learning-path scores are rule-based code written by the team	Not applicable	Not applicable

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
O*NET 31.0 Database (skills, technologies, interests, job zones)	https://www.onetcenter.org/database.html	Public	CC BY 4.0	N
DOSM and OpenDOSM labour, wage and Job Market Insights data	https://open.dosm.gov.my/data-catalogue ; https://www.dosm.gov.my	Public	OpenDOSM CC BY 4.0; DOSM releases cited as "Source: DOSM"	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
MyCOL 2024/2025 critical occupations list (TalentCorp)	https://www.talentcorp.com.my/mycol/	Public	Published report, cited	N
arXiv paper counts per category	https://export.arxiv.org/api/query	Public	arXiv metadata CC0	N
UN SDG targets	https://unstats.un.org/sdgs/indicators/indicators-list/	Public	UN, cited	N
Free course and video links (Microsoft Learn, OpenLearn, Linux Foundation, freeCodeCamp, Harvard CS50, Google, AWS, Simplilearn)	data/courses.csv and data/learning_resources.csv in the repository	Public (links and titles only)	Each creator's terms; linked, not copied	N
IT skills test (90 questions) and interest questionnaire (36 items)	data/it_assessment.json and data/interest_items.json	Self-made by the team	Apache 2.0 (project licence)	N
Demo student profiles	data/sample_profiles.json	Synthetic (fictional people)	Apache 2.0 (project licence)	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code (Anthropic, Claude Opus models)	Writing and testing code, data cleaning and checks, research for sources, drafting documentation	Major
ChatGPT (OpenAI)	Discussing ideas and answering questions in chat; not used to write code	Minor
Replit Agent (Replit)	Deployment settings on Replit	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

The team set the direction, the 6 IT areas, the questionnaire items and the score weights; Claude Code wrote code to that plan. Every change was run before merging: 140 automated tests, a data validator that checks each row has a source and date, and a walk-through of the app in a browser on desktop and phone. Data the AI gathered was checked against the original sources, and every learning link was opened. Team members reviewed the code and can explain each part; no number in the app is invented.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Cloud credits and platform (Replit Workspace and Deployments)	Development workspace and hosting of the live prototype

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit is where the live prototype runs, but the app does not depend on it: it is one Python program with no packages to install, so it runs anywhere with Python 3.12 (python src/main.py) without code changes. No sponsor GenAI API was used; the Fireworks AI key is the team's own.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

There are no accounts or logins, and we add no analytics of our own; Replit Deployments adds its standard page-view analytics script. Questionnaire and test answers and skill ratings stay in the browser tab (sessionStorage) and are sent to our server only to compute results; the server keeps nothing.

If a student uploads a CV, its text is sent to our server and, to write the summary, to Fireworks AI. Our server never writes it to disk, logs or caches it, and the page asks students to leave out their IC number, address and phone. "Explain this for me" sends only computed match facts, no personal data. The API key is kept in Replit Secrets, not in the code. Demo profiles are fictional.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Hallucination: the model only words text. A reply with any number not in our computed facts (or the CV) is rejected and a rule-based text is shown instead.

Single provider: if Fireworks AI is down or out of credit, every feature still works with the fallback text.

Data and bias: role skills come from O*NET (U.S. data). Malaysian demand data per IT role is thin (DOSM publishes job ads for only one IT occupation; MyCOL is one edition), so trend scores are coarse; every number shows its source and date, and a missing number is never estimated. Scores guide the student and never hide a role; the weights are team choices shown on screen. Learning links can change; each shows the date it was checked.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Team leader: Hong Zhi Yang Signature: 

Date: 24 October 2026